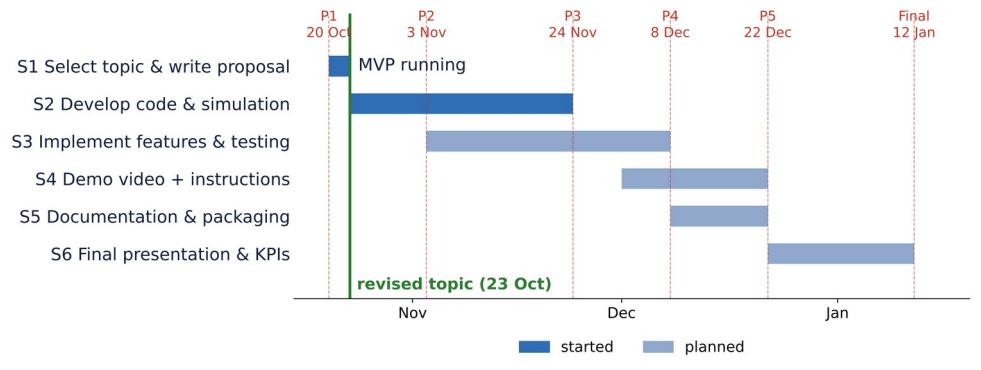
MDM2 – Case Study: Intelligent Systems in Production
One-Page Proposal

Team	Team 1
Members	Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas
Roles & Cases	Team lead and core code: Yu Han. Work packages are shared by all four members.
Project Title	Second Check of AOI Results with Supervised Machine Learning: Quality Control and Defect Detection in Electronics Assembly
GitHub Repository URL	https://github.com/bodhi584/aoi-decision-gate
Contact Email	melisa.subasi@stud.th-deg.de · han.yu@stud.th-deg.de · alejandro.rivas@stud.th-deg.de · meron.hagos@stud.th-deg.de
Industrial Application (target domain/use-case)	Quality inspection in electronics assembly, based on the lecture example "Siemens – Quality Control and Defect Detection" (p. 18). After automated optical inspection (AOI), flagged boards go to manual inspection. Current problem: on 21 Oct 2026 Siemens and NVIDIA announced an "AI second check of AOI results" in the Siemens electronics factory in Erlangen, targeting about 40 % fewer false calls.
Keywords (3–6)	quality control, defect detection, automated optical inspection, supervised machine learning, binary classification, big data processing
Submission Date (YYYY-MM-DD)	2026-10-23
Gant Chart	
1) Problem Statement & Measurable Outcomes (3–4 sentences)	After soldering, an AOI camera system checks every solder joint against fixed limits and sends flagged boards to manual inspection. In real Siemens production data about 98 % of the flagged boards are false calls, so operators check thousands of good boards. Input: the AOI alarm (measurements of a flagged board) → System: binary classification (supervised machine learning) trained on past manual-inspection results → Output: for each AOI alarm, true defect (manual inspection) or false call (skip manual inspection). Measurable outcomes: defective boards classified as false calls ≤ 1 %; false calls correctly identified ≥ 20 % (baseline 6.4 %); precision and recall.
2) Motivation & Industrial Relevance (2–3 sentences)	Key motivations: reduce dependence on manual quality checks, keep product quality, and increase efficiency of the assembly line (lecture goal: better quality). Siemens is working on exactly this problem in its Erlangen electronics factory (Oct 2026).
3) Related Work Snapshot (2–3 key references)	Pfab et al. (2024) published the Siemens AOI data. Albayari et al. (2025) used XGBoost, random forest and neural networks on the same data with a random split. Gap addressed: evaluation in time order, as on the real line, with a fixed limit on missed defects.
4) Method & Feasibility (≤6 sentences)	Methodology: CRISP-DM. Data: Siemens AOI data (Mendeley, CC BY-NC 3.0), 440,274 flagged solder joints on 39,742 boards from 132 days, about 70 measurements each, labelled by manual inspection. Models (scikit-learn): logistic regression (baseline), random forest, and a neural network with one hidden layer (lecture p. 12, 17). Simulation: the line is simulated day by day; each day is decided by a model trained on earlier days only. MVP running: in time order 1.8–6.4 % of false calls are correctly identified while 99 % of defects are caught (14–18 % with a random split). Deliverables: Python software prototype, notebook, tests on GitHub, demo video, report (IEEE template).
5) Milestones & Timeline (short table/list)	P1 23 Oct · S1 select topic & write proposal: proposal + tech stack + MVP P2 3 Nov · S2 develop code & simulation: core code + baseline sim; test plan + report P3 24 Nov · S3 implement features & testing: unit/CI status; issues board P4 8 Dec · S3 implement features & testing: full evaluation, error analysis P5 22 Dec · S4 demo video + instructions; S5 documentation & packaging Final 12 Jan · S6 final presentation & KPIs, report
6) Risks & Ethics (1–2 sentences)	Simulation-only environment; no physical machinery risk. Data anonymised by Siemens, no personal data; licence non-commercial (course use). Limits: one production line, 132 days, labels from operators. Main risk: the line changes over time, so the 1 % limit may not hold (fallback: report what works and why).